

Jiali Shan

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Education

- Ph.D. Fudan University**, School of Mathematical Sciences Shanghai, China
2025 – present
- Research areas: randomized numerical linear algebra, total least squares, backward error analysis, large-scale scientific computing.
- B.S. Fudan University**, Mathematics and Applied Mathematics Shanghai, China
2021 – 2025

Experience

- University of California, Berkeley**, Exchange Student, Department of Mathematics Berkeley, CA, USA
Aug 2024 – Dec 2024
- Coursework: all courses graded A+/A.

Publications

- Efficient estimate for the optimal backward error of the multidimensional total least squares** 2026
Jiali Shan, Yimin Wei
(SIAM Journal on Matrix Analysis and Applications, to appear)
- Optimal backward error of a total least squares and its randomized algorithms** 2025
Jiali Shan, Yimin Wei
[10.1137/25M1749657](#) (SIAM Journal on Matrix Analysis and Applications, 46(3), 2116–2139)
- Neural networks for total least squares solution of the time-varying linear systems** 2025
Xuezhong Wang, Jiali Shan, Yimin Wei
[10.1007/s40314-024-03070-1](#) (Computational and Applied Mathematics, 44, Article 106)

Talks

- Backward analysis of the total least squares and its Randomized Algorithms**
- The 8th Peking University Symposium for Outstanding PhD Students in Computational and Applied Mathematics, Peking University, Nov. 2025.
 - The 7th LuoJia Doctoral Forum in Mathematics, Wuhan University, Apr. 2026.

Funding

- National Natural Science Foundation of China (NSFC)**
- Category: Youth Student Basic Research Project
 - Role: Project leader
 - Grant: 123B1008
 - duration: 2023-2025
 - topic: Randomized Algorithms for Toeplitz Total Least Squares Problems

Awards

- 2025 ICCM Undergraduate Paper Award** Jan 2026

Skills

- Programming:** MATLAB, Python, C, LaTeX
- Languages:** Mandarin Chinese (native), English (fluent), Japanese (basic)